JEE Main 2027 — Model Question Paper 2

Duration: 3 hours · Maximum marks: 300 · 75 questions (25 each in Physics, Chemistry, Mathematics)

1. Section A of each subject: 20 multiple-choice questions, one correct option. +4 for correct, −1 for wrong, 0 if unattempted.
2. Section B of each subject: 5 numerical-value questions; enter the exact integer answer. +4 for correct, 0 otherwise (no negative marking).
3. Take g = 10 m/s² unless stated otherwise.
4. Answer key is at the end of the paper.

Physics

Section A — Multiple Choice (single correct)

1. A car takes a turn of radius 100 m on a flat road with μ = 0.4 (g = 10 m/s²). The maximum safe speed is

(1) 40 m/s **(2)** 20 m/s
(3) 10 m/s **(4)** 14 m/s

2. A 2 kg block on a rough floor (μ = 0.5) is pulled by a horizontal force of 20 N (g = 10 m/s²). Its acceleration is

(1) 5 m/s² **(2)** 10 m/s²
(3) 2.5 m/s² **(4)** 0

3. In an Atwood machine, masses 3 kg and 1 kg hang over a light frictionless pulley (g = 10 m/s²). The string tension is

(1) 30 N **(2)** 20 N
(3) 15 N **(4)** 7.5 N

4. Work done in stretching a spring (k = 200 N/m) from natural length by 10 cm is

(1) 10 J **(2)** 2 J
(3) 0.5 J **(4)** 1 J

5. The moment of inertia of a thin uniform rod of mass M and length L about an axis through one end, perpendicular to it, is

(1) $ML^2/12$ **(2)** $ML^2/3$
(3) $ML^2/2$ **(4)** ML^2

6. An ice skater spinning freely pulls in her arms so her moment of inertia halves. Her rotational kinetic energy

(1) doubles **(2)** halves
(3) stays the same **(4)** becomes four times

7. The acceleration due to gravity at a height equal to the Earth's radius is

(1) g/2 **(2)** g/√2
(3) g/4 **(4)** g/8

8. If the radius of a small rain drop falling through air is doubled, its terminal velocity becomes

(1) 2 times **(2)** √2 times
(3) 8 times **(4)** 4 times

9. Water flows through a horizontal pipe whose cross-sections are in the ratio $A_1 : A_2 = 2 : 1$. The speeds $v_1 : v_2$ are

(1) 2 : 1 **(2)** 1 : 2
(3) 1 : 4 **(4)** 4 : 1

10. If the length of a simple pendulum is increased by 21%, its period increases by

(1) 10% **(2)** 21%
(3) 11% **(4)** 42%

11. A pipe closed at one end has length 0.5 m. If the speed of sound is 340 m/s, its fundamental frequency is

(1) 340 Hz **(2)** 680 Hz
(3) 170 Hz **(4)** 85 Hz

12. Two tuning forks of 256 Hz and 260 Hz are sounded together. The number of beats per second is

(1) 2 **(2)** 8
(3) 516 **(4)** 4

13. A point charge q is at the centre of a cube. The electric flux through one face is

(1) q/ε_0 **(2)** $q/6\varepsilon_0$
(3) $6q/\varepsilon_0$ **(4)** $q/4\varepsilon_0$

14. Four charges q each sit at the corners of a square of side a. The potential at the centre is ($k = 1/4\pi\varepsilon_0$)

(1) $4\sqrt{2}\ kq/a$ **(2)** $4kq/a$
(3) $2\sqrt{2}\ kq/a$ **(4)** 0

15. A cell of emf E and internal resistance r delivers maximum power to an external resistor R when

(1) $R = 2r$ **(2)** $R = r/2$
(3) $R = r$ **(4)** $R \to \infty$

16. To convert a galvanometer into an ammeter, one connects

(1) a high resistance in series **(2)** a low resistance in series
(3) a high resistance in parallel **(4)** a low resistance in parallel

17. A 1 m rod moves at 5 m/s perpendicular to a uniform field of 0.2 T, with rod, velocity and field mutually perpendicular. The induced emf is

(1) 0.1 V **(2)** 1 V
(3) 10 V **(4)** 0.5 V

18. An ideal step-up transformer with turns ratio 1 : 10 is fed 220 V ac. The output voltage is

(1) 2200 V **(2)** 22 V
(3) 220 V **(4)** 2420 V

19. The critical angle for glass of refractive index $\sqrt{2}$ is

(1) 30° **(2)** 60°
(3) 45° **(4)** 90°

20. In Young's double slit experiment, the slit separation is halved and the screen distance doubled. The fringe width becomes

(1) unchanged **(2)** 2 times
(3) ½ times **(4)** 4 times

Section B — Numerical Value

21. The kinetic energy of a body increases by 300%. The percentage increase in its momentum is

Answer: _____

22. The resistance (in Ω) of a bulb rated 100 W, 220 V is

Answer: _____

23. A body falls freely from rest from a height of 20 m ($g = 10\ m/s^2$). Its speed (in m/s) on reaching the ground is

Answer: _____

24. ^{238}U decays to ^{206}Pb through a series of α and β decays. The number of α-particles emitted is

Answer: _____

25. 2 mol of a monatomic ideal gas is heated at constant volume from 27 °C to 127 °C. Heat supplied in J ($R = 8.3\ J\ mol^{-1}\ K^{-1}$) is

Answer: _____

Chemistry

Section A — Multiple Choice (single correct)

26. The number of atoms in 0.5 mol of N_2 is

(1) 3.011×10^{23} **(2)** 6.022×10^{23}
(3) 1.204×10^{24} **(4)** 1.5×10^{23}

27. The maximum number of electrons in the shell with n = 3 is

(1) 8 **(2)** 9
(3) 18 **(4)** 32

28. Which molecule has the largest dipole moment?

(1) NH_3 **(2)** NF_3
(3) BF_3 **(4)** CO_2

29. The shape of ClF_3 is

(1) trigonal planar **(2)** pyramidal
(3) see-saw **(4)** T-shaped

30. The number of atoms per unit cell in a face-centred cubic lattice is

(1) 2 **(2)** 4
(3) 1 **(4)** 6

31. The van der Waals constant 'a' accounts for

(1) intermolecular attraction **(2)** molecular volume
(3) molecular mass **(4)** kinetic energy

32. For an exothermic reaction at equilibrium, raising the temperature

(1) shifts it forward **(2)** has no effect
(3) shifts it backward **(4)** increases K

33. For an acetic acid–sodium acetate buffer, pH = pKa when

(1) [salt] = 10 [acid] **(2)** [acid] = 10 [salt]
(3) [acid] = 0 **(4)** [salt] = [acid]

34. Entropy increases in

(1) freezing of water **(2)** melting of ice

(3) condensation of steam **(4)** crystallisation of sugar

35. Which 0.1 M aqueous solution has the highest boiling point?

(1) $Al_2(SO_4)_3$ **(2)** NaCl

(3) glucose **(4)** urea

36. Which undergoes SN1 reaction fastest?

(1) CH_3Br **(2)** CH_3CH_2Br

(3) $(CH_3)_3CBr$ **(4)** $(CH_3)_2CHBr$

37. Hofmann bromamide degradation of acetamide gives

(1) ethylamine **(2)** acetic acid

(3) methanol **(4)** methylamine

38. Which undergoes aldol condensation?

(1) HCHO **(2)** CH_3CHO

(3) C_6H_5CHO **(4)** $(CH_3)_3CCHO$

39. Which undergoes the Cannizzaro reaction?

(1) C_6H_5CHO **(2)** CH_3CHO

(3) CH_3COCH_3 **(4)** CH_3CH_2CHO

40. The strongest base in aqueous solution is

(1) NH_3 **(2)** CH_3NH_2

(3) $(CH_3)_2NH$ **(4)** $C_6H_5NH_2$

41. The most electronegative element is

(1) O **(2)** Cl

(3) N **(4)** F

42. The number of three-centre two-electron (3c–2e) bonds in diborane is

(1) 1 **(2)** 2

(3) 4 **(4)** 0

43. Ziegler–Natta catalyst is

(1) $TiCl_4 + Al(C_2H_5)_3$ **(2)** $Pd/BaSO_4$

(3) Ni **(4)** V_2O_5

44. The catalyst used in Haber's process is

(1) Pt **(2)** V_2O_5

(3) Fe **(4)** Ni

45. The number of peptide bonds in a tripeptide is

(1) 3 **(2)** 1

(3) 0 **(4)** 2

Section B — Numerical Value

46. Moles of O_2 required for complete combustion of 4 mol of CH_4 is

Answer: _____

47. The pH of a solution with $[H^+] = 10^{-3}$ M is

Answer: _____

48. The number of σ-bonds in ethene (C_2H_4) is

Answer: _____

49. Given $C + O_2 \rightarrow CO_2$, $\Delta H = -393$ kJ and $CO + \frac{1}{2}O_2 \rightarrow CO_2$, $\Delta H = -283$ kJ. The magnitude of ΔH (kJ) for $C + \frac{1}{2}O_2 \rightarrow CO$ is

Answer: _____

50. The oxidation number of Mn in $KMnO_4$ is

Answer: _____

Mathematics

Section A — Multiple Choice (single correct)

51. If $\sin\theta + \cos\theta = \sqrt{2}$, then $\sin\theta\cos\theta$ is

(1) 1 **(2)** 1/2
(3) $\sqrt{2}$ **(4)** 0

52. The number of solutions of $\sin x = 1/2$ in $[0, 2\pi]$ is

(1) 1 **(2)** 4
(3) 2 **(4)** 0

53. The sum of the first 20 terms of the AP 3, 7, 11, … is

(1) 820 **(2)** 800
(3) 860 **(4)** 780

54. If ${}^nC_2 = 45$, then n is

(1) 9 **(2)** 11
(3) 15 **(4)** 10

55. The middle term of $(x + 1/x)^{10}$ is

(1) 210 **(2)** 252
(3) 120 **(4)** 126

56. The circle with centre (1, 2) passing through the origin is

(1) $x^2 + y^2 - 2x - 4y = 0$ **(2)** $x^2 + y^2 + 2x + 4y = 0$
(3) $x^2 + y^2 - 2x - 4y = 5$ **(4)** $x^2 + y^2 - x - 2y = 0$

57. The eccentricity of the hyperbola $x^2/16 - y^2/9 = 1$ is

(1) 3/4 **(2)** 5/3
(3) 5/4 **(4)** 4/5

58. The slope of the tangent to $y = x^3$ at $x = 2$ is

(1) 8 **(2)** 6
(3) 4 **(4)** 12

59. The maximum value of $f(x) = x(4 - x)$ is

(1) 2 **(2)** 4
(3) 8 **(4)** 16

60. $\int e^x(\sin x + \cos x)\,dx$ equals

(1) $e^x \sin x + C$ **(2)** $e^x \cos x + C$
(3) $-e^x \sin x + C$ **(4)** $e^x(\sin x - \cos x) + C$

61. $\int_0^1 x\, e^x\, dx$ equals

(1) e **(2)** e − 1

(3) 1 **(4)** 2e − 1

62. $\lim_{x\to 0} (1 - \cos x)/x^2$ equals

(1) 0 **(2)** 1

(3) 2 **(4)** 1/2

63. The function f(x) = |x| at x = 0 is

(1) discontinuous **(2)** continuous but not differentiable

(3) differentiable **(4)** neither continuous nor differentiable

64. The order and degree of $(d^2y/dx^2)^3 + (dy/dx)^4 = 0$ are

(1) 2, 3 **(2)** 2, 4

(3) 3, 2 **(4)** 1, 4

65. A and B are independent with P(A) = 0.5 and P(B) = 0.4. P(A ∪ B) is

(1) 0.9 **(2)** 0.2

(3) 0.7 **(4)** 0.8

66. The area of the triangle with vertices (0, 0), (4, 0), (0, 3) is

(1) 12 **(2)** 7

(3) 5 **(4)** 6

67. The direction cosines of a line equally inclined to the positive coordinate axes are

(1) 1, 1, 1 **(2)** 1/√3, 1/√3, 1/√3

(3) 1/3, 1/3, 1/3 **(4)** 1/√2, 1/√2, 0

68. The distance between the planes x + 2y + 2z = 3 and x + 2y + 2z = 9 is

(1) 2 **(2)** 6

(3) 3 **(4)** 4

69. The contrapositive of p → q is

(1) q → p **(2)** ~p → ~q

(3) ~q → ~p **(4)** p → ~q

70. The inverse of the matrix [[2, 1], [1, 1]] is

(1) [[1, 1], [1, 2]] **(2)** [[2, −1], [−1, 1]]

(3) [[−1, 1], [1, −2]] **(4)** [[1, −1], [−1, 2]]

Section B — Numerical Value

71. If $\log_2 x + \log_2 (x - 2) = 3$, then x is

Answer: _____

72. The number of diagonals of a decagon is

Answer: _____

73. The remainder when 2^{100} is divided by 3 is

Answer: _____

74. If $f(x) = x^2 + 2x$ and g(x) = x + 1, then f(g(1)) is

Answer: _____

75. The number of three-digit numbers divisible by 7 is

Answer: _____

Answer Key

Physics

1. (2)	**2.** (1)	**3.** (3)	**4.** (4)	**5.** (2)	**6.** (1)
7. (3)	**8.** (4)	**9.** (2)	**10.** (1)	**11.** (3)	**12.** (4)
13. (2)	**14.** (1)	**15.** (3)	**16.** (4)	**17.** (2)	**18.** (1)
19. (3)	**20.** (4)	**21.** 100	**22.** 484	**23.** 20	**24.** 8
25. 2490					

Chemistry

26. (2)	**27.** (3)	**28.** (1)	**29.** (4)	**30.** (2)	**31.** (1)
32. (3)	**33.** (4)	**34.** (2)	**35.** (1)	**36.** (3)	**37.** (4)
38. (2)	**39.** (1)	**40.** (3)	**41.** (4)	**42.** (2)	**43.** (1)
44. (3)	**45.** (4)	**46.** 8	**47.** 3	**48.** 5	**49.** 110
50. 7					

Mathematics

51. (2)	**52.** (3)	**53.** (1)	**54.** (4)	**55.** (2)	**56.** (1)
57. (3)	**58.** (4)	**59.** (2)	**60.** (1)	**61.** (3)	**62.** (4)
63. (2)	**64.** (1)	**65.** (3)	**66.** (4)	**67.** (2)	**68.** (1)
69. (3)	**70.** (4)	**71.** 4	**72.** 35	**73.** 1	**74.** 8
75. 128					

Original model paper prepared for practice, following the latest announced exam pattern. Not an official NTA paper.